

Active Removal of Space Debris : A Feasibility Analysis

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Abstract—Since the space age began, rocket launches have crowded Earth’s orbit with satellites, causing a growing space debris issue. Orbital Debris Removal Mission offers a promising solution to reduce debris and maintain sustainable space operations. This paper presents a drag augmentation method using expanded foam to increase the area-to-mass ratio of debris, thereby enhancing atmospheric drag to speed up orbital decay. The system, which uses a docking mechanism to apply the foam, enables debris to undergo uncontrolled re-entry and burn up in the atmosphere. Combined with electric propulsion, this approach is designed to allow a satellite to spray ten large debris objects within a year, and cause their deorbit in 25 years.

Index Terms—Satellite, Space Debris, Removal

I. INTRODUCTION

A. Background

As of 2024, the issue of space debris has reached a critical point. Since the launch of the first satellite in 1957, the number of objects in Earth’s orbit has increased significantly. Currently, over 12,000 satellites have been deployed, with more than 7,500 still operational. In 2023 alone, more than 2,800 satellites were launched into Low Earth Orbit (LEO), causing significant congestion in altitudes between 500–600 km. This growing population of active satellites and space debris presents heightened risks for collisions, particularly in LEO, where the vast majority of satellites and space debris resides shown in figures 1 and 2.

The increasing frequency of events such as anti-satellite weapon tests and rocket body explosions has substantially increased the problem. Initiatives such as the European Space Agency’s (ESA) ”Zero Debris Charter” [1] aim to address these risks through controlled reentries and Active Debris Removal (ADR) missions. These efforts emphasize the need for innovative and sustainable methods of space debris mitigation to ensure the continued viability of space operations.

B. Motivation for the study

In order to combat this growing issue this study aims to investigate the feasibility of a low-cost, sustainable ADR

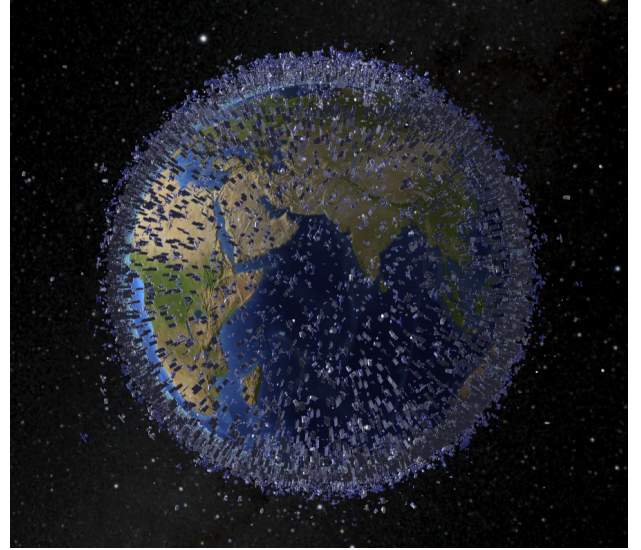


Figure 1: Space debris population in the LEO region [2]

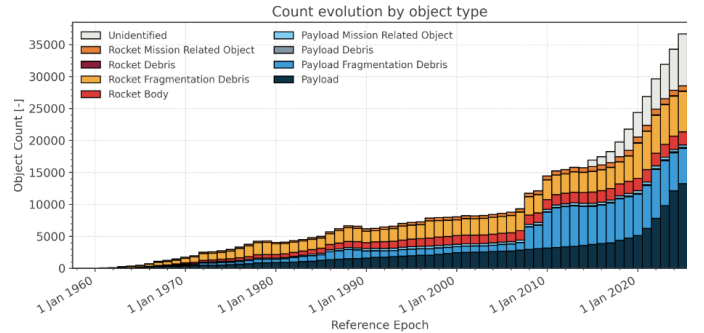


Figure 2: Increase in space debris over time [3]

mission. This study was done to give insight into the main drivers of an ADR mission and to understand what areas can be improved in order to make future ADR missions more promising. However the main purpose of this study is to perform a preliminary mission analysis of a LEO satellite system, and to plan a simple space mission on a conceptual level.

C. Motivation for selected active debris removal technology & technology readiness level

Currently, the foam-based approach is at a Technology Readiness Level (TRL) of 4-5 which indicates that the concept has been validated in a lab environment and some components have been tested in a relevant environment. However, full-scale testing in space is still needed before it can be fully operational. The method has shown promise in small-scale tests and simulations but requires further refinement to ensure the foam can reliably adhere to various types of debris and expand consistently in space conditions [4].

Compared to other ADR technologies, such as robotic arms, nets, harpoons, the foam method stands out due to its relatively low complexity and reduced operational demands [5]. Traditional methods often require precise maneuvering, complex docking systems and significant energy consumption. In contrast, the foam system allows for a simpler "spray-and-go" approach, where the debris is left to naturally decay over time without the need for extensive post-deployment intervention.

Given the potential of this technology, its further development could significantly lower the cost and complexity of debris removal missions. As the TRL progresses toward levels 6-7, where systems are demonstrated in relevant operational environments, we expect increased interest from space agencies and private operators in integrating this method into future debris mitigation strategies.

D. Limitations, simplifications and General assumptions

Given that this study is a preliminary feasibility analysis of a foam-based debris removal method, several simplifying assumptions were made to streamline the calculations and scope:

- i. The satellite's trajectory is assumed to be unaffected by external forces such as atmospheric drag (until mission completion), solar radiation, or gravitational influences from celestial bodies other than Earth.
- ii. All debris orbits are considered circular due to their low eccentricity values ($e \leq 0.06$).
- iii. The debris is modeled as cooperative, meaning no angular momentum is assumed, which simplifies the docking phase between the satellite and debris.
- iv. Low-thrust maneuvers are considered without factoring in atmospheric drag or the Earth's oblateness, and the argument of perigee shift and Right Ascension of the Ascending Node (RAAN) variations over time are neglected.

- v. The foam's expansion is assumed to be uniform and ideal, maintaining its structure without deformation under external forces.

These assumptions allow for a manageable scope while providing insight into the method's effectiveness in a simplified environment.

E. Structure

The paper consists of an introduction, methods, results, discussion and conclusion. The methods section contains the rocket, selection of targets, launch site, rocket simulation, the foam deorbiting system and cost estimation. The results section, presents rocket simulation, orbital maneuvers and the cost. The discussion analyzes the choice of the launch vehicle, earth satellite operations, cost implications and sustainability concerns regarding rocket launches and the importance of satellites. The conclusions summarize the study's findings, calling for further research to optimize foam technology and improving satellite design and electric propulsion systems to support active debris removal efforts.

II. METHODS

A. Rocket

The goal is to launch a satellite with a mass of roughly 2,500 kg to a Low Earth Orbit (LEO) with an appropriate rocket while minimizing the cost of the mission. To do so, the SpaceX Falcon 9 Block 5 was chosen for its competitive price and proven track record. Additionally the rocket is reusable as the first stage and fairings can be recovered, which lowers the environmental impact. Since the satellite mass is 2,500 kg, and Falcon 9 has a payload capacity of 22,800 kg to LEO [6], the satellite is expected to be launched as part of a SpaceX Rideshare program to reduce cost and waste. This also means that the rocket's entire payload capacity likely will be filled.

Table I: Falcon 9 v1.2 specifications

Type	Value
Total mass	549,054 kg
Total payload to LEO	22,800 kg
Structural mass of the first stage	25,600 kg
Propellant mass of the first stage	395,700 kg
Structural mass of the second stage	3,900 kg
Propellant mass of the second stage	92,670 kg
Average thrust of the first stage	7917 kN
Average specific impulse of the first stage	296.5 s
Thrust of the second stage	981 kN
Specific impulse of the second stage	348 s

B. Selection of targets

The primary goal of this project is to actively remove space debris from Low Earth Orbit. Focusing specifically on a set of derelict rocket boosters. For this purpose, a total of ten old rocket boosters, originally part of the Thor launch system from the 1970s, were selected as optimal targets for de-orbiting. Several criteria were used in the selection process to ensure the efficiency and success of the mission.

1) *Orbital Proximity*: By targeting objects with similar orbits, it is possible to minimize the time and fuel required for maneuvering between them, thereby optimizing mission efficiency. The important parameters that were analysed was the inclination, RAAN, apogee and perigee of the orbits. According to [7] there are three areas where the amounts of debris is at a critical level, which should initially be considered. The region considered in this paper is the one around the Sun-Synchronous Orbit(SSO). The SSO is one of the most populated regions for commercial and scientific purposes which also contains large amounts of space debris in it. The parameters for SSO are

- Inclination between 97 and 100 degrees
- Altitude between 600 and 900 km
- Eccentricity smaller than 0.035

This results in a list of 140 possible targets. To make the mission fuel efficient ideally the targets should be in the same orbit. That is not realistic, therefore the velocity changes that are required to change the orbits from one target to the next one was calculated, denoted ΔV .

For inclination change (i)

$$\Delta V_i = V \sqrt{2 - 2\cos(\frac{\pi}{2} \Delta i)} \quad (1)$$

For Semi-major axis (a)

$$\Delta V_a = \left| \sqrt{\frac{\mu}{a_0}} - \sqrt{\frac{\mu}{a_1}} \right| \quad (2)$$

For RAAN (Ω)

$$\Delta V_\Omega = \frac{\pi}{2} \sqrt{\frac{\mu}{a}} \left| \Delta \Omega \right| \sin(i) \quad (3)$$

A closer look at these equations shows that a change in RAAN and inclination are the most expensive maneuvers in terms of ΔV . Because the targets are in polar orbit with only a maximum of 3 degrees in Δi . Therefore the RAAN of the orbits becomes the limiting parameter. Because the RAAN changes overtime makes it very hard to predict the RAAN on mission day when the spacecraft will intercept that specific debris. This report will not make a extensive analysis of the RAAN. The values where chosen and approximated within a

span of 100 degrees for the worst case. Table II shows all the orbital parameters of the targets that was chosen with the closest RAAN. All the targets was from the same type except one to make the mission simpler and to make the foam more effective.

Table II: Table of chosen targets and orbital parameters

SAT-id	Mass(kg)	Perigee(km)	Apogee(km)	RAAN(°)	i (°)
2940	65	816.4	869.4	109.8	98
4048	65	762.7	823.5	110	98
3271	65	804.8	878.8	110	98
7412	65	788.4	853.5	112	99
7817	115	795.0	873.1	114	99
2741	65	765.8	840.7	117	99
7219	65	767.6	858.5	124	99
4954	65	742.6	794.3	126	98
6788	65	789.4	831	130	99
4332	65	751.6	836.1	130	99

C. Launch site

The chosen launch site is Cape Canaveral Air Force Station (CCAFS) in Florida, specifically Space Launch Complex 40 (SLC-40). This site is well-known for hosting launches by SpaceX. The exact coordinates of the launch complex are 28.562106°N and 80.577180°W.

We will be following a trajectory similar to that of the OneWeb L17 mission, which successfully launched 40 satellites on March 9, 2023 shown in figure 3. During this mission, the rocket reached a near Sun-Synchronous Orbit (SSO) at an altitude of 600 km and an inclination of 87°. This type of orbit is ideal for our satellite's mission objectives [8].

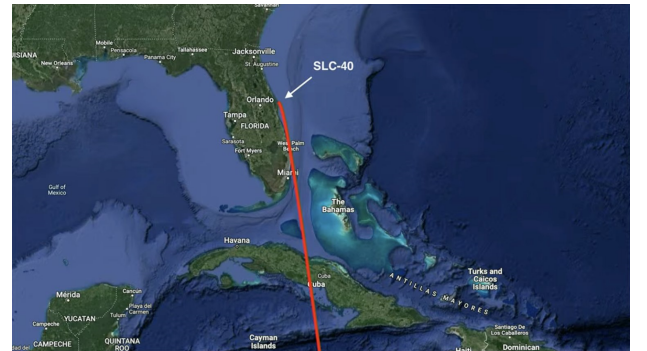


Figure 3: Rocket path of the OneWeb L17 launch on March 9, 2023.[8]

Cape Canaveral Air Force Station is one of the best launch sites because it is only 28 degrees from the equator and provides an extra velocity boost from Earth's rotation, which means lower fuel requirements to reach orbit and a rise in payload capacity. Our site location will still allow for

homeport launch efficiency, although our mission targets a Sun-Synchronous Orbit (SSO). It also helps eliminate scheduling conflicts, as launching eastward over the Atlantic Ocean provides clear safe zones downrange, with miles of open water available to manage air and marine traffic during launches that might shed debris and help ensure none falls on populated areas.

CCAFS also provides strong infrastructure for launch operations. The site is also well-connected to transportation networks, facilitating the movement of equipment and personnel, making it ideal for preparing and executing complex space missions. These combined factors make CCAFS a strong choice for our mission.

D. Rocket simulation

The rocket launch is simulated using the values in Table I. The target altitude is selected to be the altitude of the perigee of the first target space debris, 816.4 km and an inclination of 98° (Table II). The first stage is modeled to use 70% of its fuel-reserve to thrust the rocket, leaving 30% as a fuel-reserve for it to land on the Landing Zone 1 of Cape Canaveral.

To simulate the rocket launch, several models were used:

1) *Gravity*: The acceleration due to gravity is simulated using Newtons Law of Universal Gravitation.

$$g = \frac{G \cdot M}{(R_E + H)^2} \quad (4)$$

where:

- g is the gravitational acceleration (m/s^2),
- G is the gravitational constant ($6.67430 \times 10^{-11} \text{ N} \cdot \text{m}^2 \cdot \text{kg}^{-2}$),
- M is the mass of the Earth ($5.972 \times 10^{24} \text{ kg}$),
- R_E is the Earth's radius (approximately 6,371 km),
- H is the height above the Earth's surface (m).

2) *Atmospheric density*: The density of the atmosphere was simulated by pre-calculating the density at 100 m intervals using a separate model for each layer of the atmosphere: troposphere, tropopause, stratosphere, upper stratosphere, stratosphere, mesosphere, mesopause and thermosphere. The temperature and pressure at each altitude was used to calculate the density of the air using the Ideal Gas Law.

$$\rho = \frac{P}{R_{\text{gas}}} \cdot T \quad (5)$$

where:

- ρ is the air density (kg/m^3),
- P is the pressure (Pa),
- R_{gas} is the specific gas constant for dry air ($287.05 \text{ J/(kg} \cdot \text{K)}$),

- T is the absolute temperature (K).

3) *Air drag*: The drag force (D) acting on an object moving through a fluid, in this case air, is given by the equation:

$$D = \frac{1}{2} \cdot \rho \cdot V^2 \cdot C_d \cdot A \quad (6)$$

where:

- D is the drag force (N),
- ρ is the fluid density (kg/m^3),
- V is the velocity of the object relative to the fluid (m/s),
- C_d is the drag coefficient (dimensionless), which depends on the shape of the object,
- A is the cross-sectional area of the object perpendicular to the flow (m^2).

The rocket goes through two distinct stages in the launch: the sounding stage and the gravity turn stage.

1) *Sounding rocket stage*: During the first stage of the launch, the rocket behaves as a sounding rocket. This means it's oriented vertically and is subjected to thrust, drag and gravity. The equations of motions can be described as follows:

$$\begin{cases} m \frac{dV}{dt} = T - D - W \\ \frac{dH}{dt} = V \end{cases} \quad (7)$$

where:

- $T = -V_e \frac{dm}{dt}$ is the thrust,
- $D = \frac{1}{2} \rho A v^2 C_d$ is the drag,
- $W = mg$ is the weight.

2) *Gravity turn stage*: When the altitude reaches around 100 m, the gravity turn is initialized by introducing a minor pitch angle which causes a tiny part of the gravitational force to be directed perpendicular to the longitudinal axis. This angle must remain tiny while the pressure exerted on the rocket is large, since rockets are structurally weaker in the transverse direction, so maintaining low kinetic pressure is necessary when the angle of attack is not zero to prevent structural failure. By expressing the equations of motion in the local-horizon frame, a set of coupled non-linear ordinary differential equations can be obtained, which describe the dynamics of the rocket's motion accurately.

$$\begin{cases} \dot{V} = \frac{T}{m} - \frac{D}{m} - \left(g - \frac{\dot{X}^2}{R_e + H} \right) \sin \gamma \\ \dot{\gamma} = -\frac{1}{V} \left(g - \frac{\dot{X}^2}{R_e + H} \right) \cos \gamma \\ \dot{X} = V \cos \gamma \\ \dot{Y} = V \sin \gamma \\ \dot{m} = -\beta(t) \end{cases} \quad (8)$$

E. Satellite

Considering the total weight of the subsystems, such as: Telemetry, Sensors, Foam Spray System, CDH, AGN, ADCS, Thermal Control, Antenna, Power System, Fuel Tanks, Drag Sail and the whole structure, a conservative estimation of the dry mass is around 800 kg.

The power necessary to maintain all the satellite systems and thrusters operational is approximately 21kW of continuous power, which will be provided by 45m² of QJ Solar Cell 4G32C, with an efficiency of about 32% [9]. A contingency battery that will be able to provide power to the sub-systems will also be used, a lithium battery of 5kW/h was chosen with a specific energy density of 170 Wh/kg, which weights about 35kg.

1) *De-orbiting system:* In this project the Active Space Debris Removal System that was chosen is an expanding foam system. Which is a passive de-orbiting method that has the goal to increase the area-to-mass ratio, so that their natural re-entry is favored. This method is a new alternative to the usual debris removal system, and therefore has not yet been optimized shown in figure 4.

A polymeric matrix and an expanding gas (blowing agent) were chosen to implement the foam method, through the use of conservative assumptions based on the current state of the art technology. A polyurethane based foam with an expanded density

$$\rho_{\text{foam}} = 10 \text{ kg/m}^3$$

and with a foam expansion factor of 2:1, to have a better adherence to the debris and avoiding the nucleation of the foam in the wrong way. The application of such foam is obtained through the use of a foam ejecting nozzle, mounted on a robotic arm that grants full control over the sprayed surface of targeted debris, this choice was based on the comparison of the other possible methods, which resulted in the nozzle as it has the higher reliability (7), a technological feasibility (8), a flexibility (6) and cost/effectiveness (7), on a scale from 1 to 10.

In order to have an optimal drag contribution that allows the debris to de-orbit within 25 years, a few parameters must be followed, where the most important one was the area-to-mass ratio

$$\frac{A}{m} = \frac{\pi r^2}{\frac{4}{3} \rho_F \pi r^3 + m_D} \quad (9)$$

which, according to the literature, must be in our case, bigger than 0.07, since the chosen debris orbits are between

700 km and 900 km, and the other one is the optimal radius shown in figures 5 and 6, [10].

$$r_{\text{opt}} = \sqrt[3]{\frac{3}{2} \cdot \frac{m_D}{\pi \rho_F}} \quad (10)$$

In order to remove from the equations the variables that could influence the de-orbiting time of the debris we have increased the minimum area to mass ratio, which will grant us a shorter de-orbiting time. This resulted in a total foam mass of approximately 700 kg from table III.

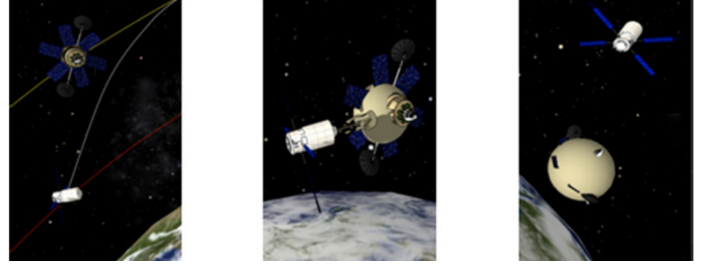


Figure 4: Representation of the proposed method: target debris interception (left), foaming process (center), debris deorbiting (right). [10]

Table III: In this table we can see the Optimal Radius (O.R), the Mass Of Foam (M.O.F) and Optimal Volume (O.V)

SAT-id	Mass (kg)	O.R(m)	O.V(m ³)	M.O.F(kg)	A/m(m ² /kg)
2940	65	1.4587	13	65	3.4278
4048	65	1.4587	13	65	3.4278
3271	65	1.4587	13	65	3.4278
7412	65	1.4587	13	65	3.4278
7817	115	1.7642	23	115	3.4278
2741	65	1.4587	13	65	3.4278
7219	65	1.4587	13	65	3.4278
4954	65	1.4587	13	65	3.4278
6788	65	1.4587	13	65	3.4278
4332	65	1.4587	13	65	3.4278

2) *Propulsion system:* For Earth close missions, where the optimal specific impulse required is between 3000s and 3500s, the best compromise considering both the performance and cost, is for sure the Hall effect electric engine, visualized in figure 7. The electric propulsion schemes are mainly based on the acceleration of charged ions by means of electromagnetic forces, a gas is used as propellant, in our case Xenon will be used, since its easy to ionized, its high atomic number provide an high specific impulse and its easy to store.

The choice of the specific thruster falls on the PPS-5000 family, produced by Safran, with an nominal operative power of 5kW, a specific impulse of 3000s and an efficiency larger than 50%. This results in over 250mN of thrust per single PPS-5000, conservatively. Since the required time for finishing the

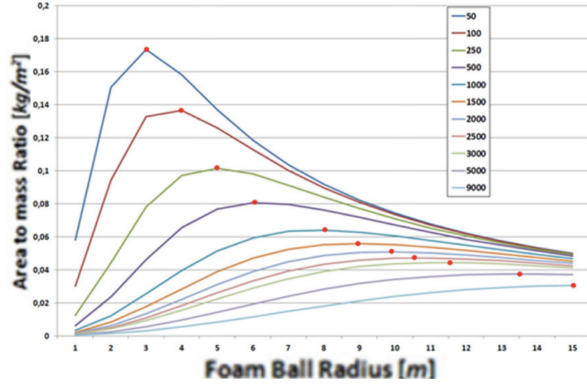


Figure 5: Area-to-mass ratio vs. foam ball radius for different values of the initial debris mass[10]

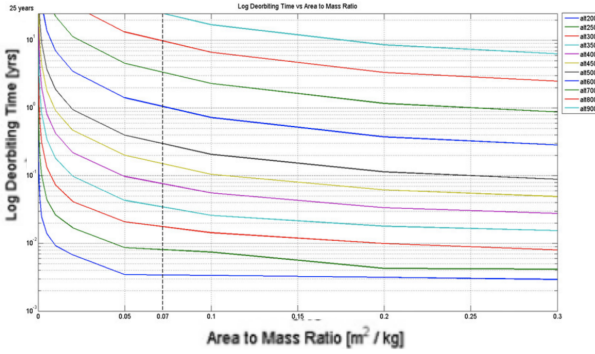


Figure 6: Deorbiting time vs. area-to-mass ratio [10]

de-orbiting of the 10 debris is fixed to 1 year, the necessity to increase the thrusts resulted in mounting on board of our satellite 4 of these thrusters, with a resulting maximum thrust of 1.1-1.2N, even though for most of the mission output will be fixed at 70-80% of the maximum value.

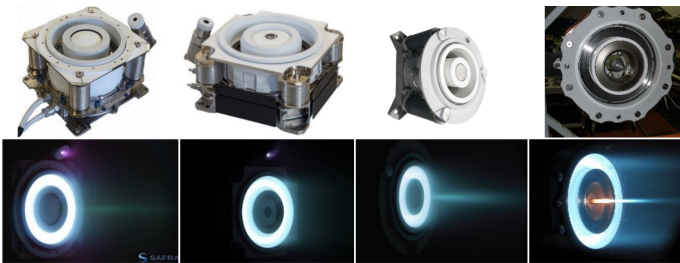


Figure 7: Safran Hall thrusters, from left to right: the commercially available PPS®1350 and PPS®5000; the PPS®X00 under active development; and the PPS®20k technology demonstrator [11]

3) *Satellite de orbiting*: Once all the debris have been foamed, the purpose of this vessel have been reached, therefore

is necessary to dispose of it like any other debris. The method we have choose for the de-orbiting is a drag sail of 55 m², which will grant the destruction of the satellite within 25 years 8 in the Earth's atmosphere. The material used for the sail is Kapton, very resistant to the harsh environment of space.

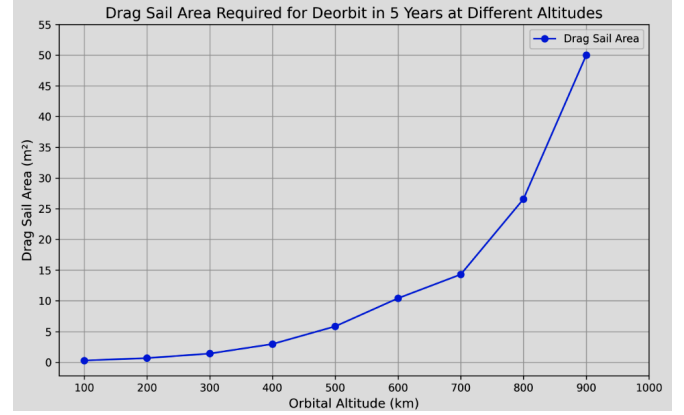


Figure 8: Drag Sail Area vs Altitude for deorbit in 20 years [12]

F. Cost analysis

The feasibility of the mission is greatly dependent on its economical viability, which made it important to minimize costs. In order to accomplish this, different Two-Stage-To-Orbit (TSTO) commercial launchers were compared in terms of launch cost and payload capability in order to determine the cheapest option shown in table IV.

Table IV: Launch Vehicles Price and Derived Price per Kilogram to Low-Earth Orbit [13]

Rocket	Launch Cost (million \$)	Range of capacity to LEO (kg)	Range of price per kilogram (\$)	Reusable
Delta IV Medium/Heavy	210-513	9,420-28,790	17,840-22,360	NO
Atlas V	144-220	8,210-18,850	12,220-21,660	NO
Antares	103-109	6,200-6,600	16,500-17,600	NO
H-IIA/B	115-144	10,000-16,500	8,760-11,560	NO
Long March 2D	38	3,500	11,000	NO
Ariane 5 ECA	229	21,000	10,890	NO
Ariane 64	126	21,500	5,860	NO
Falcon 9	70	22,800	3070	YES

The costs for the satellite were estimated for each subsystem based from its weight and associated cost per kg where applicable which can be seen in table VI. The power system was similarly estimated using the required power generation instead of weight. Additionally costs for ground control, system integration, verification and insurance are added.

III. RESULTS

A. Rocket

Figure 9 displays the evolution of the different rocket parameters (velocity, flight path angle and altitude) as a function of time.

In order to stay in the circular orbit targeted at 816.4 km altitude, the rocket need to have a particular speed:

$$V = \sqrt{\frac{G \cdot M}{R_E + H}} \approx 7.447 \text{ km/s} \quad \text{when } H = 816.4 \text{ m}$$

This speed matches the final speed obtained in the simulation at the specified altitude, as seen in Figure 9.

The results can also be plotted as altitude as a function of downrange distance, as seen in Figure 10.

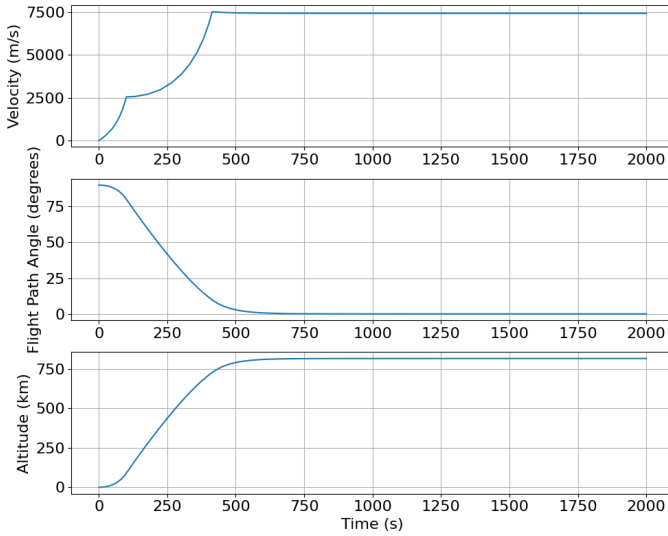


Figure 9: Rocket velocity, flight path angle and altitude as a function of time.

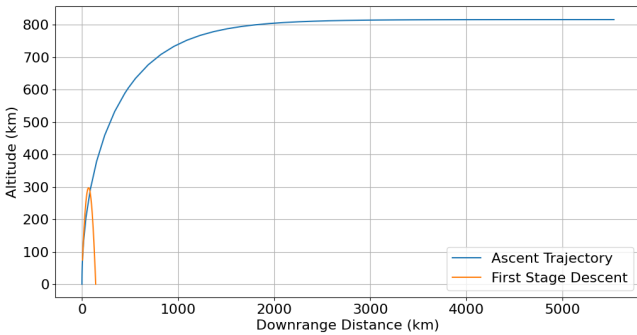


Figure 10: Rocket altitude as a function of downrange distance.

B. Orbital maneuvers

In order to perform all the necessary maneuvers, the total ΔV for each transfer and intercept has been estimated in

17.5917 km/s which resulted in a duration of the mission of 279.5 days, at which we have added a contingency 70 days if unforeseen events should present, leading to a total mission duration (intended as the time necessary to foam all the targeted debris) of 349 days. The ΔV that was calculated is shown in table V and the specific ΔV for each of the orbital maneuvers. The resulting fuel consumption is about 1100 kg, considering the contingency. The ΔV for the first debris is zero because the satellite lands in that orbit with the rocket booster so no fuel from the satellite is used.

Table V: Different ΔV for changing orbital parameters and duration for each maneuvers in days

SAT-id	ΔV_a	ΔV_{Ω}	ΔV_i	$\Delta V_{tot}(\text{km/s})$	$\Delta t(\text{days})$
2940	0	0	0	0	0
4048	0.6723	0.4901	0.1151	1.2775	30.6753
3271	0.6581	0.3180	0.0138	0.9900	22.3843
7412	0.2746	0.7754	0.1438	1.1938	25.2009
7817	0.1737	1.2615	0.0565	1.4917	29.0427
2741	0.4152	1.6984	0.0995	2.2130	38.6787
7219	0.1347	4.4319	0.3072	4.8738	69.5887
4954	0.6335	1.3710	0.0149	2.0194	25.7385
6788	0.5946	2.2039	0.1831	2.9816	32.6513
4332	0.2272	0.2107	0.1130	0.5509	5.5821
Total	3.7839	12.7609	1.0469	17.5917	279.5425

C. Cost Analysis

The price for the subsystems/components of the satellite can be seen in table VI. Accounting for system integration and verification at an estimated cost of \$1.0 million each the cost of the satellite is estimated to be \$27.5 million including an insurance cost of 10%. Ground control was estimated to cost between \$3 to \$6 million. Launch costs were estimated to range from \$7.5 to \$11.0 million. The total mission cost was projected to be approximately \$38.0 to \$44.5 million

IV. DISCUSSION

A. Rocket

The SpaceX Falcon 9 Block 5 was chosen for its proven track record and competitive cost per kilogram when launched using the SpaceX Rideshare program. The cost is an important factor but one must also take into account the reliability when choosing a launch provider. At the time of writing, the Falcon 9 is a proven reliable rocket with a mission success rate of 99.23% [14].

The results of the simulated gravity turn accent indicate that the SpaceX Falcon 9 Block 5 is capable of performing the maneuvers necessary to place the satellite payload in the desired orbit altitude. However, since the launch is planned

Table VI: Cost of subsystems/components

Subsystem/Component	Weight (kg)	Cost (\$ per kg)	Cost (million \$)
Antenna	N/A	N/A	0.5
Altitude and control	80	10,000	0.8
Chassis	160	9,000	1.4
Foam	700	10	0.007
Foam housing	N/A	N/A	0.15
Foam pump	N/A	N/A	0.1
Guidance and navigation	48	5,000	0.24
Optical and radar sensors	N/A	N/A	3.0
Power	N/A	N/A	6.3
Propellant	1000	3,000	3.0
Propulsion	N/A	N/A	6.0
Robotic arm	N/A	N/A	1.0
Spray mozzle	N/A	N/A	0.05
Telemetry	48	5,000	0.24
Thermal	32	6,000	0.19

to take place as a part of a SpaceX Rideshare mission, the total mission feasibility will depend on the mass of the other satellite payloads in conjunction with the desired target altitudes of the other payloads.

B. Earth satellite operations

From the results we can see that starting from satellite n° 2940 (where the second stage of the rocket leaves us), the major contribution to our ΔV is due to the change of RAAN, therefore we choose targets with similar values of this parameter which resulted in lower fuel consumption (and transfer time).

The total duration of the mission is calculated at 279.5 days, excluding the time needed to intercept and spray foam on the debris. In the worst-case scenario, we allocate up to 7 days for each debris to account for the time needed to maneuver, intercept, and spray the foam effectively. Given this timeline, even with multiple debris targets, the mission will remain under one year in duration, aligning with our initial goal.

By factoring in these operational contingencies, such as the 7-day window for each debris, we ensure flexibility while still adhering to the overall mission timeline.

Regarding the use of foam, we have to highlight that since it is still a method of de-orbiting under study, and therefore has not reached the state of art level, targets had to be chosen in a manner where there was an upper limit to the weight in order to avoid the necessity to bring on board large masses of foam. In the future, when better kinds of foam have been developed, it will be possible to ease those parameters that currently limit the employment of this method (mass of foam to transport and adherence on the target).

C. Cost analysis

As of 2024, the cost to launch a Falcon 9 Block 5 rocket has risen to approximately \$69.75 million [14], which is a slight increase from the previous price due to inflation and rising material costs. This standard Falcon 9 Block 5 launch can carry payloads up to 22,800 kg into low-Earth orbit, with a cost-per-kilogram of around \$3,000–\$4,400. This will result in approximately \$7.5–\$11.0 million in costs for the launch. Regarding the satellite the cost was calculated per subsystem/component when applicable. The satellite chassis is made from aluminum and carbon-fiber reinforced polymer to ensure the satellite is lightweight and durable. Typically the cost for the chassis is around \$3,000–\$12,000 per kg where a higher cost of \$9,000 per kg was chosen to account for design complexity. Similarly other costs for the subsystems were chosen to be on the higher end due to the mission's complexity and need for high grade subsystems/components. The propulsion system uses four PPS-5000 Hall-effect thrusters at an estimated \$1.5 million per unit, totalling \$6.0 million. The propellant system uses 1000 kg of xenon gas at an estimated cost of \$3,000 per kg leading to a total cost of \$3.0 million for the propellant system. The power systems were estimated to cost \$6.3 million. This cost was derived from the need for 21 kW of continuous power generation which was estimated at a cost of around \$300,000 per kW of power for space grade, high efficiency solar panels. The foam deorbiting system was estimated to cost \$3.5 million. The cost was based on the cost of the spray nozzle, foam pumping system, foam and robotic arm. Due to the low TRL of the foam system it will also have to be verified and validated and thus undergo extensive testing which will increase the costs [15]. The insurance cost was added onto the development and manufacturing costs of the satellite and was set to 10% due to the low TRL and high complexity of the mission. The costs for ground control operations were estimated to cost between \$3 to \$6 million. Summing up the costs for the launch, satellite development and the ongoing operational ground control, the total mission is estimated to cost approximately \$38.0 to \$44.5 million. The proposed budget is competitive when compared to other active space debris removal initiatives, especially given the novel foam-based de-orbiting technology, while the launch cost is to be split between the missions that the rocket will bring in orbit based on payload weight.

D. Sustainability

Comparing the energy cost between a rocket launch and a long distance flight gives a clear image as to how big the environmental impact of rocket launches are. The Falcon 9

rocket has a RP-1 tank capacity of 155,800 kg where the specific energy of RP-1 is 12 kWh/kg which gives an energy cost of 1.9 GWh [6]. Compared to a long-distance flight with a Boeing-747 spanning ten hours using approximately 150,000 litres of RP-1 the energy cost is 1.5 GWh. Thus the energy cost for a rocket launch is approximately 27% more expensive than that of a long-distance airplane flight.

On a global scale the environmental impact from rocket launches is very small. Certainly it contributes to global warming but its emissions are simply a fraction to that of the aviation industry which itself is small on the global scale. One of the most prominent environmental impacts from rocket launches is its contribution to the depletion of the ozone layer. Their impact on the ozone layer is unique in the fact that rockets are the only man-made object which directly distributes ozone-destroying chemicals into the stratosphere for which even small amounts can greatly affect the ozone layer. With decreasing launching costs and increased interest in the space sector the amount of rocket launches is going to substantially increase. While the effects from launches are currently small this could very easily become a major cause for ozone depletion. In order to maintain the ozone levels this could become a limiting factor for the allowed amount of future launches and of importance for future space industry [16]. The local environmental impact is much more noticeable as opposed to the global impact. Rocket launches can potentially change the local ecology and soil quality. Especially from the launch byproduct of hydrochloric acid spread through the ground cloud which drastically reduces the pH levels in the soil and nearby water surfaces for a short duration, which have killed local fish populations. The effects from the nitrogen oxides which also form during launch can result in eutrophication in water bodies and ecosystem acidification [17]. For unsuccessful launches there will be high levels of unburned toxic propellant left in the environment.

Satellites have become an integral part of our daily lives. The technology has a wide variety of use cases such as Global Positioning System (GPS) navigation, weather forecasting, communication, television and radio broadcasting. Satellites are also used for environmental monitoring to combat deforestation, illegal fishing and other issues which need to be constantly observed. Furthermore it plays an integral role for studying and understanding climate change, such as monitoring air quality and monitoring the ice caps [18]. Another area of research in which satellites play a major role is that of outer space, where satellites allow for more clear capturing of data as opposed to a telescope down on earth.

V. CONCLUSION

This study has demonstrated the feasibility of using foam-based methods to enhance the removal of space debris in LEO. By employing Hall effect electric engines and an expanding foam system, this approach aims to deorbit large debris objects, specifically old rocket boosters, by increasing their area-to-mass ratio and accelerating their orbital decay. The satellite system, with a total mass of 2,500 kg and a propulsion configuration utilizing four PPS-5000 thrusters, is designed to complete the mission within one year.

The results of this study are promising, but further research is required to optimize the foam technology and refine the satellite's design for operational use. Future work should focus on improving the adherence of foam to irregularly shaped debris and reducing the amount of foam required per target. Additionally, continued development of electric propulsion systems could further reduce mission costs and extend operational lifetimes, making active debris removal more viable on a larger scale.

VI. DIVISION OF WORK

Carl Bergholtz - Paper writing, Debris/Satellite; Brian Bertacco - Paper writing, Debris/Satellite/Cost Analysis; Alexander Robson - Paper writing, Rocket; Corentin Gauthé - Paper writing, Introduction; Peter Ohlson - Paper writing, Cost Analysis/Sustainability.

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